# CO1-AT2-Short Answers

**Course Code: ITA0329** **Name: V. Naga Nischay**

**Course name: Mobile Computing** **Regno:192472255**

## Q3.

**Given:**

Number of carriers = 100

Time slots per carrier = 8

**Formula:** Total voice channels = Number of carriers × Time slots per carrier

**Solution:**

= 100 × 8

= 800

**Answer:** 800 voice channels

## Q4.

**Given:**

Base station power = 80 W

Handset power = 4 W

**Formula:** Power ratio = $10 \log(P_1 / P_2)$

**Solution:**

= $10 \log_{10}(80 / 4)$

= $10 \log_{10}(20)$

≈ 13.01 dB

**Answer:** 13.01 dB

## Q5.

**Given:**

Carrier power = −60

Interference power = −92

**Formula:** C/I = Carrier power − Interference power

**Solution:**

= −60 − (−92)

= 32 dB

**Answer:** 32 dB

## Q9.

**Given:**

Cluster size = 4

Channels per cell = 15

**Formula:** Total channels = Cluster size × Channels per cell

**Solution:**

= 4 × 15

= 60

**Answer:** 60 channels

## Q11.

**Given:**

Frame duration = 5

Number of slots = 8

**Formula:** Slot duration = Frame duration ÷ Number of slots

**Solution:**

= 5 ÷ 8

= 0.625 ms

**Answer:** 0.625 ms

## Q12.

**Given:**

Total bandwidth = 24 MHz = 24,000

Carrier spacing = 200 kHz

**Formula:** Number of carriers = Total bandwidth ÷ Carrier spacing

**Solution:**

= 24,000 ÷ 200

= 120

**Answer:** 120 carriers

## Q15.

**Given:**

Number of cells = 6

Calls per cell = 250

**Formula:** Total capacity = Number of cells × Calls per cell

**Solution:**

= 6 × 250

= 1,500

**Answer:** 1,500 simultaneous calls

## Q17.

**Given:**

Number of channels = 150

Bandwidth per channel = 25 kHz

**Formula:** Total bandwidth = Number of channels × Bandwidth per channel

**Solution:**

= 150 × 25

= 3,750 kHz = 3.75 MHz

**Answer:** 3.75 MHz

## Q18.

**Given:**

Number of frequency bands = 30

Guard-band width = 40 kHz

Number of gaps between 30 bands = 30 − 1 = 29

**Formula:** Total guard bandwidth = Number of gaps × Guard-band width

**Solution:**

= 29 × 40

= 1,160 kHz = 1.16 MHz

**Answer:** 1.16 MHz

## Q20.

**Given:**

Regional centers = 8

Base stations per center = 40

Users per base station = 250

**Formula:** Total users = Centers × Base stations per center × Users per base station

**Solution:**

= 8 × 40 × 250

= 80,000

**Answer:** 80,000 users

## Q22.

**Given:**

Total bandwidth = 6 MHz = 6,000 kHz

Bandwidth per channel = 30 kHz

**Formula:** Number of channels = Total bandwidth ÷ Bandwidth per channel

**Solution:**

= 6,000 ÷ 30

= 200

**Answer:** 200 channels

## Q24.

**Given:**

Total bandwidth = 40 MHz = 40,000 kHz

Bandwidth per user = 400 kHz

**Formula:** Maximum users = Total bandwidth ÷ Bandwidth per user

**Solution:**

= 40,000 ÷ 400

= 100

**Answer:** 100 simultaneous users

## Q25.

**Given:**

Number of base stations = 1,200

Users per base station = 600

**Formula:** Total capacity = Number of base stations × Users per base station

**Solution:**

= 1,200 × 600

= 720,000

**Answer:** 720,000 simultaneous users

## Q26.

**Given:**

Total cells = 200

Cells paged per round = 4

**Formula:** Paging rounds = Total cells ÷ Cells paged per round

**Solution:**

= 200 ÷ 4

= 50

**Answer:** 50 rounds

## Q28.

**Given:**

Total subscribers = 600,000

Roaming percentage = 15%

**Formula:** Roaming subscribers = (Percentage ÷ 100) × Total subscribers

**Solution:**

= (15 ÷ 100) × 600,000

= 90,000

**Answer:** 90,000 subscribers

## Q29.

**Given:**

Maximum capacity = 12,000 calls/s

Actual load = 9,000 calls/s

**Formula:** Utilization = (Actual load ÷ Maximum capacity) × 100

**Solution:**

= (9,000 ÷ 12,000) × 100

= 75%

**Answer:** 75%

## Q32.

**Given:**

Total handoff attempts = 2,500

Success rate = 99%

**Formula:** Successful handoffs = (Success rate ÷ 100) × Total attempts

**Solution:**

= (99 ÷ 100) × 2,500

= 2,475

**Answer:** 2,475 successful handoffs

## Q34.

**Given:**

Number of cells = 7

Users per cell = 900

**Formula:** Total capacity = Number of cells × Users per cell

**Solution:**

= 7 × 900

= 6,300

**Answer:** 6,300 simultaneous users

## Q35.

**Given:**

Frame duration = 4.8 ms

Number of slots = 8

**Formula:** Slot duration = Frame duration ÷ Number of slots

**Solution:**

= 4.8 ÷ 8

= 0.6 ms

**Answer:** 0.6 ms

## Q38.

**Given:**

Number of cells = 9

Users per cell = 600

**Formula:** Total capacity = Number of cells × Users per cell

**Solution:**

= 9 × 600

= 5,400

**Answer:** 5,400 users

## Q39.

**Given:**

Total calls per second = 60,000

Dropped-call percentage = 0.4%

**Formula:** Dropped calls = (Dropped percentage ÷ 100) × Total calls

**Solution:**

= (0.4 ÷ 100) × 60,000

= 240

**Answer:** 240 dropped calls per second

## Q41.

**Given:**

Coverage per station = 2.5 km²

Number of stations = 150

**Formula:** Total coverage = Coverage per station × Number of stations

**Solution:**

= 2.5 × 150

= 375 km²

**Answer:** 375 km²

## Q43.

**Given:**

Total subscribers = 150,000

Users processed per cycle = 6,000

**Formula:** Paging cycles = Total subscribers ÷ Users per cycle

**Solution:**

= 150,000 ÷ 6,000

= 25

**Answer:** 25 cycles

## Q45.

**Given:**

Signal power = −85 dBm

Noise power = −115 dBm

**Formula:** SNR (dB) = Signal power − Noise power

**Solution:**

= −85 − (−115)

= 30 dB

**Answer:** 30 dB

## Q48.

**Given:**

Number of control channels = 250

Bandwidth per channel = 12 kHz

**Formula:** Total bandwidth = Number of channels × Bandwidth per channel

**Solution:**

= 250 × 12

= 3,000 kHz = 3 MHz

**Answer:** 3 MHz